



Index Number:

NATIONAL CERTIFICATE OF EDUCATION

2024

TECHNOLOGY STUDIES (N550)

TIME: 1 HOUR 15 MINUTES

Component 1

Candidates answer on the Question Paper.
Additional materials: Standard drawing equipment

READ THESE INSTRUCTIONS FIRST

- Write your index number in the space provided above and on the A3 printed question sheet.
- Write in dark blue or black ink. Coloured and HB pencils may be used for diagrams/sketches and shading.
- Answer **ALL** questions.
- Check whether this assessment booklet consists of:
 - Questions **1 - 7** printed on **8** pages from page **2** to page **9**;
 - Question **8** printed on an **A3** sheet of paper as an insert.
- Do not use correction fluid, highlighters or glue.
- All answers must be written in the spaces provided.
- The total marks for each question is given at the beginning of each question.
- The number of marks for each item is shown in brackets [].
- Question 8:**
 - All drawings must be accurately drawn using drawing instruments.
 - All construction lines **MUST** be shown.
 - All dimensions are in millimetres.
- The total number of marks for this paper is **50**.
- At the end of the examination, fold the A3 sheet and insert it in the assessment booklet.**

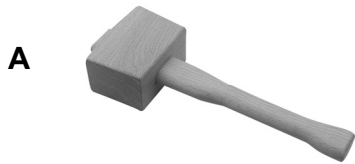
For Examiners' use											
Question No.	1	2	3	4	5	6	7	8(a)	8(b)	Total	Signature
Examiner											
Team Leader											
CE/ACE											

SECTION A

Question 1 (5 marks)

Circle the correct answer. Each item carries one mark.

1. Which of the following shows a **measuring tape**?



2. Which tool is shown in **Figure 1**?



Figure 1

A An engineer's square

B A bench hook

C A try square

D A sliding bevel

3. Which of the following is an example of a thermoplastic?

A Rigid Polyurethane

B Polyvinyl Chloride

C Polyester resin

D Urea formaldehyde

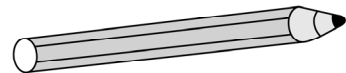
4. Which of the following is an example of a holding tool?

A



Scriber

B



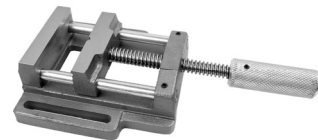
Pencil

C



Hack saw

D



Machine vice

5. Which of the following eco-green labels represents 'products containing plastics that can be recycled'?

A



B



C



D



Question 2 (5 marks)

Put a tick (✓) in the appropriate column to indicate whether the statements in the table below are **True** or **False**. An example is given.

Statement		True	False
Example: Biodegradable plastic bags are examples of eco-materials.		✓	
a	High carbon steel is a non-ferrous metal.		
b	A hammer is a driving tool.		
c	Bamboo stems are examples of eco-materials.		
d	A steel rule is a holding tool.		
e	Red cedar is a softwood.		

E

TL

CE

Question 3 (5 marks)

Fill in the blanks with the appropriate word(s) from the list given below. An example is given. There is one extra word in the list.

Acrylic	Chipboard	Pine
Copper	Teak	
Epoxy resins	Pressurised air	

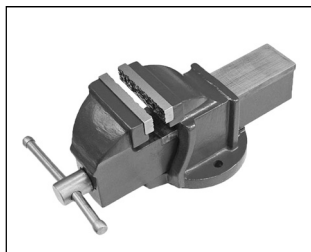
Example: Epoxy resins have good adhesive qualities.

- a) is a hardwood.
- b) is a thermoplastic.
- c) is a good conductor of heat and electricity.
- d) is a manufactured board.
- e) is used in pneumatic systems.

Question 4 (5 marks)

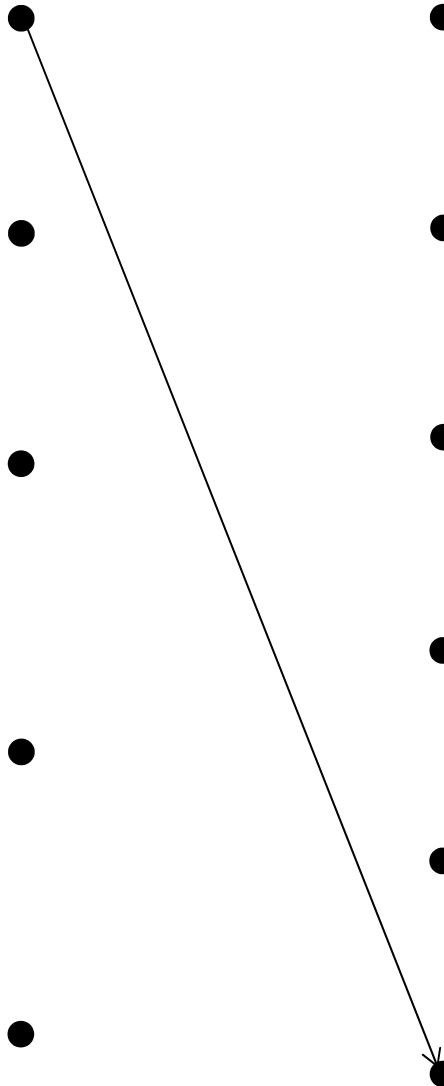
Match the tools shown in **Column A** to their names in **Column B**. An example is given. There is one extra word in **Column B**.

Column A



Column B

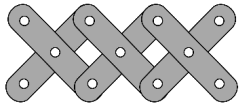
- Jack plane
- Engineer's vice
- Chisel
- Tenon saw
- Hand drill
- **Screw driver**
- Snip



Question 5 (6 marks)

- (a) Name each mechanical component in the table below. Choose from the given list. An example is given.
There is one extra word in this list.

Gear	Pulley	Chain and sprocket
Linkage	Cam	

Mechanical component	Name
Example: 	 Linkage
	
	
	

[3]

- (b) Sketch a **hand saw** in the space provided below.

[3]

SECTION B**Question 6 (4 marks)**

Figure 2 shows an isometric drawing of a shaped block.

Apply tone shading techniques to the shaped block using pencils and/or coloured pencils.
The arrow shows the direction of light.

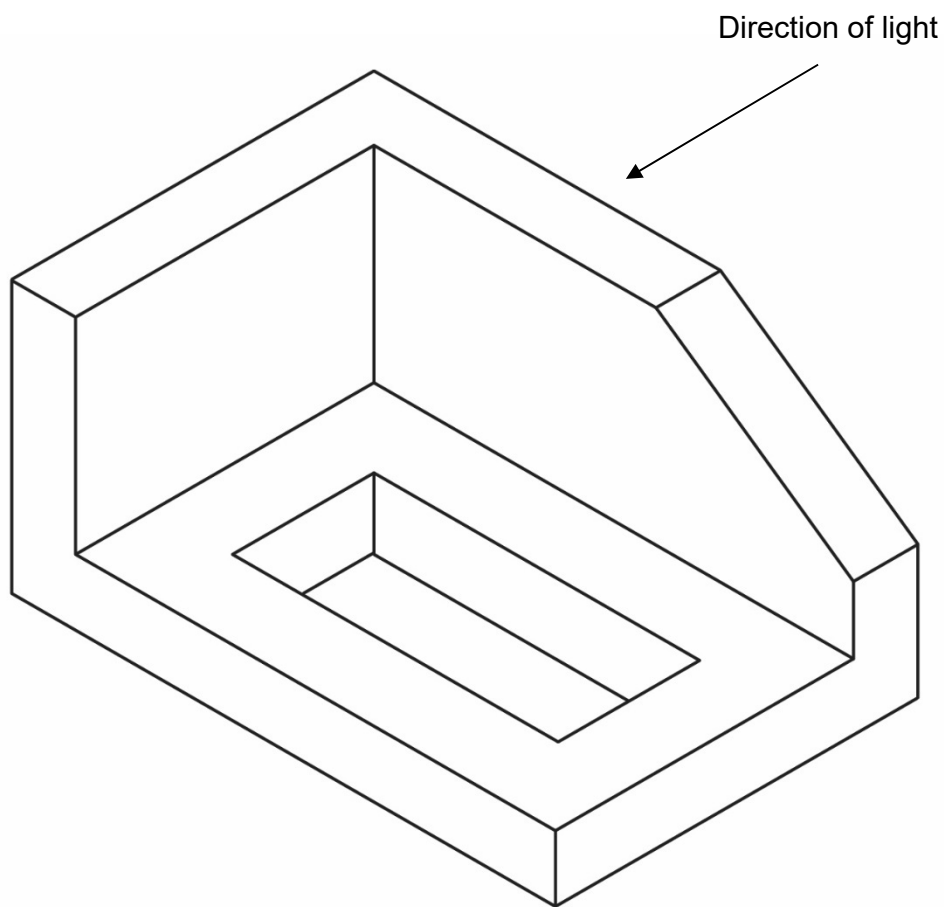


Figure 2: Isometric drawing of a shaped block

E

TL

CE

Question 7 (5 marks)

Figure 3 shows a set of three spice jars. A spice rack is required to store the spice jars safely in the kitchen.



Figure 3: A set of spice jars

In the space provided, sketch **one** idea for the spice rack to hold the three jars and complete the analysis for the material(s) used and its safety.

Material(s) used:	Safety:
.....	
.....	

For Question 8
Please see A3 insert

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